The FLAIR Privacy Boundary

Code crosses rightward · only aggregate JSONs return · patient data does not.

PUBLIC SIDE · researcher · code · aggregator · leaderboard

BARRIER

12+ INSTITUTIONAL SILOS · data stays here · no internet · no root · immutable image

DEVELOP — train against MIMIC-CLIF

Researcher

MIMIC-CLIF
~70k stays · public

FLAIR CLI

`flair load-task` → cohort + splits

train model → y_true · y_pred · y_prob

`flair build-report` → 4 aggregate JSONs

TRIPOD+AI-compliant outputs

Container
model + code · frozen

code in

containerised code · S

same bytes execute at e

PRIVACY BOUNDARY

patient data never crosses

`table1` `tripod` `fairnesscurves`
4 JSONs

aggregate JSONs only · cell-suppressio

aggregates out

Site 1 — CLIF tables · local — Data steward — HIPAA

Site 2 — CLIF tables · local — Data steward — HIPAA

Site 3 — CLIF tables · local — Data steward — HIPAA

Site 4 — CLIF tables · local — Data steward — HIPAA

Site 5 — CLIF tables · local — Data steward — HIPAA

Site 6 — CLIF tables · local — Data steward — HIPAA

Site 7 — CLIF tables · local — Data steward — HIPAA

Site 8 — CLIF tables · local — Data steward — HIPAA

Site 9 — CLIF tables · local — Data steward — HIPAA

Site 10 — CLIF tables · local — Data steward — HIPAA

Site 11 — CLIF tables · local — Data steward — HIPAA

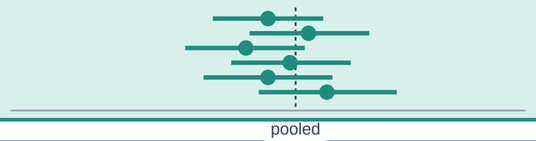
Site 12+ — CLIF tables · local — Data steward — HIPAA

each silo: container runs in CLIF-WorkBench; only aggregate JSONs leave the site

AGGREGATE — central verification

FLAIR aggregator
verifies SHA-256 chain matches across sites · merges aggregates

PUBLISH — public leaderboard



Per-site AUROC
forest plot
95% bootstrap CI
variance is the signal

pooled
0.84
[0.79–0.88]

T1 · ICU mortality

T2 · LTACH discharge

T3 · Extubation failure

T4 · Sepsis (Abx-anchored)

T5 · AKI (KDIGO-2)

+ by community PR

H HIPAA-safe
cell-suppressed at n<10

catalog grows by GitHub pull request · acceptance via MIMIC-CLIF reproducibility + clinical review + site-governance review